Tilara Polyplast Pvt. Ltd.

Doc.No: QC -011/15

Rev.No. 001.13.6.2020

Techinacal Report - Polycarbonate Sheet

Sr.No.	Properties	Test Method	Unit	Value
GP)	**General Properties**			
1	Density	ISO 1183	gm/cm3	1.2
2	Light Transmission (Depending On Thickness)	ASTM D 1003	%	81-90%
3	Refractive Index	ISO 489		1585
4	Water Absorbption	ISO 62	mg	8
5	UV Protective			114
MP)	**Mechanical Properties**			
1	Tensile Strangth at Yield	ISO - 527-2	Mpa	60
2	Tensile Modulus	ISO - 527-2	Mpa	2300
3	Elongation at break	ISO - 527-2	%	>100
4	Elongation at Yield		%	6
5	Flexural Modulus	ISO - 178	Mpa	2330
6	Charpy Un - Notched	ISO - 179	kJ/m2	Not Break
7	Izod Impact notched	ISO - 180a	kJ/m2	>65
8	Flexural Stength		N/mm2	80
9	Rockwell Hardness		R	>110

TP)	**Thermal Properties**			
1	Temp of deflection (HDT) underload of 1.8MPa	ISO - 75-1	℃	130
2	Vicat Softening temp (50° C/h 50n)	ISO - 306	℃	144
3	Thermal Conductivity	DIN52612	W/m x ℃	0.2
4	Coefficient of linear		mm/(mx1/℃)	0.07
5	GWFI (Glow-wireflammability Index)	IEC60695-2	℃	900
EP)	**Electical Properties**			
1	Volume Resistivity	IES60093	Ω x cm	3x10 /14
2	Surface resistivity , Dry	IES60093	Ω	6x10/16
3	Die Electric Strength, Short time 3.2mm	D149	KV/MM	15
4	Die electric strength on 1mm Thick Specimen	DIN 53482	KV/MM	30
5	Die electric constant At 50hz	D150/din 53483		3
6	C.T.I (Tracking Index)	IEC 112		200
7	Dissipation Factor 1Mhz	IES60250		0.009
8	Dissipation Factor 100Hz	IES60250		0.0006
FP)	**Flammable Properties**			
1	Flammability rating for thickness Above 3.5mm			UL 94V0
2	Flammability rating for thickness Above 3.5mm and below			UL 94V2
MP)	**Machinability Properties**			
1	Cut,Bend and Drill.			

TILARA POLYPLAST PVT.LTD.